

# Tracking Neutron Stars Evolution: Period and Period Derivative Changes Over Time.

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## Abstract

This homework-paper presents a detailed analysis of the evolution of the period  $P$  and its time derivative  $\dot{P}$  for neutron stars modeled as vacuum orthogonal rotators. By integrating the period over time, we examine the dynamics of the system under different magnetic field conditions. Specifically, we analyze three scenarios: a constant magnetic field with  $B_d = 10^{13}\text{G}$  and  $B_d = 5 \times 10^{14}\text{G}$ , as well as an evolving magnetic field  $B_d = 5 \times 10^{14}\text{G} \cdot \exp(-t/\tau)$  with  $\tau = 10,000\text{yrs}$ . The effect of an initial period of  $P_0 = 3\text{ms}$  is also investigated. Results are presented through analytical formulas and plots that illustrate the trajectory in the  $P - \dot{P}$  diagram over a timespan of up to 1 million years. The study aims to provide insights into the long-term behavior of neutron stars under varying magnetic field conditions.

## 1 Analytical Solutions and Stability Analysis

We start by considering the electromagnetic spin-down equation for a neutron star:

$$P\dot{P} = KB_d^2 \quad (1)$$

where

$$\begin{aligned} K &= f_\chi \frac{2\pi^2}{3} \frac{R_*^6}{I_{c3}} = \\ &= 2.44 \times 10^{-40} f_\chi \frac{R_6^6}{I_{45}} \text{s} \cdot \text{G}^{-2} \end{aligned} \quad (2)$$

with  $R_6 = \frac{R_*}{10^6\text{cm}}$ ,  $I_{45} = 1$ ,  $R_6 = 1$  and  $f_\chi = 1$ , it is possible to figure out the time-evolution analytical solution for the (1). Casting into (1) the conditions (2) we get

$$K = 2.44 \times 10^{-40} \text{s} \cdot \text{G}^{-2}. \quad (3)$$

The general solution, from (1), is gotten as follow

$$P \frac{dP}{dt} = KB_d^2;$$

upon integrating, we obtain

$$\int_{P_{in}}^{P_{fin}} P dP = \int_{t_{in}}^{t_{fin}} K B_d^2(t') dt'.$$

Assuming  $K$  is constant (under the conditions of equation (2)), we simplify this to:

$$\frac{P^2}{2} \Big|_{P_{in}}^{P_{fin}} = \int_{t_{in}}^{t_{fin}} K B_d^2(t') dt'.$$

Thus, the period evolution becomes

$$P^2 = P_0^2 + 2K \int_0^t B_d^2(t') dt',$$

where the boundary conditions are  $P_{in} = P_0$ ,  $P_{fin} = P$  and  $t_{in} = 0$ ,  $t_{fin} = t$ . The resulting quadratic equation for  $P$  provides two possible solutions, with the physically meaningful one being the positive root. To ensure the physical validity of the chosen parameters, we calculate the star's mass  $M_\star$  using the following relation for the moment of inertia

$$M_\star \simeq \frac{5}{2} \frac{I}{R_\star^2}$$

for  $I = 10^{45} \text{ g} \cdot \text{cm}^2$  and  $R_\star = 10^6 \text{ cm}$ , this yields

$$\begin{aligned} M_\star &\simeq \frac{5}{2} \frac{10^{45}}{10^{12}} \text{ g} \simeq \\ &\simeq \left(2 + \frac{1}{2}\right) \times 10^{33} \text{ g} \simeq \\ &\simeq \left(2 \times 10^{33} + \frac{1}{2} \times 10^{33}\right) \text{ g} \simeq \\ &\simeq M_\odot + \frac{1}{4} M_\odot \simeq \\ &\simeq 1.25 M_\odot, \end{aligned}$$

where  $M_\odot$  is the solar mass. The maximum angular velocity  $\omega_{max}$  is given by:

$$\omega_1^{max} = \left( \frac{GM_\star}{R^3} \right)^{\frac{1}{2}} \simeq 12\,915.7 \frac{\text{rad}}{\text{s}};$$

and the corresponding minimum period is

$$P_1^{max} = \frac{2\pi}{\omega_1^{max}} \simeq 0.000486477 \text{ s} \simeq 0.486 \text{ ms} < P_0.$$

This result ensures the model's stability, as it indicates that the neutron star will not collapse due to centrifugal forces.

We also discuss the upper limit for the period, considering the decay of the magnetic field and the constraints of detectability. As a reference, we compare with the latest observations of the slowest neutron star, ASKAPJ1839-0756, observed in 2024, which has a period of  $23221.740 \pm 0.332\text{s}$ , and an angular velocity of  $2.70 \times 10^{-4} \frac{\text{rad}}{\text{s}}$  [1]. Although this observation does not provide a definitive upper limit, it offers insight into the potential values that can be achieved.

## 2 Case Studies: Varying Magnetic Fields

Next, we present the analysis of three different cases for varying magnetic field strengths, assuming an initial period  $P_0 = 0.1\text{s}$ .

1.  $B_d = 10^{13}\text{G}$ ,

$$\begin{aligned} P_1^2(t) &= P_0^2 + 2KB_d^2t = \\ &= (0.1\text{s})^2 + 2(2.44 \times 10^{-40}\text{sG}^{-2})t(10^{13})^2\text{G}^2 = \\ &= 0.01\text{s}^2 + (4.88 \times 10^{-14}t)\text{s}. \end{aligned}$$

Thus we get the period

$$P_1(t) = [0.01\text{s}^2 + (4.88 \times 10^{-14}t)\text{s}]^{\frac{1}{2}}, \quad (4)$$

therefore the angular velocity of the star in function of time

$$\omega_1(t) = 2\pi [0.01\text{s}^2 + (4.88 \times 10^{-14}t)\text{s}]^{-\frac{1}{2}}, \quad (5)$$

and the period derivative

$$\dot{P}_1(t) = \frac{4.88 \times 10^{-14}\text{s}}{2[0.01\text{s}^2 + (4.88 \times 10^{-14}t)\text{s}]^{\frac{1}{2}}} \quad (6)$$

The procedure is the same for all others cases.

2.  $B_d = 5 \times 10^{14}\text{G}$ .

$$\begin{aligned} P_2^2(t) &= (0.01\text{s}^2) + (4.88 \times 10^{-40}\text{sG}^{-2})(5 \times 10^{14})^2\text{G}^2t = \\ &= (0.01\text{s}^2) + (1.22 \times 10^{-10})t. \end{aligned}$$

Thus the period

$$P_2(t) = [1.22 \times 10^{-10}t + 0.01]^{\frac{1}{2}} \quad (7)$$

The angular velocity

$$\omega_2(t) = 2\pi [(0.01\text{s}^2) + (1.22 \times 10^{-10})t]^{-\frac{1}{2}}, \quad (8)$$

and the period derivative

$$\dot{P}_2(t) = \frac{6.1 \times 10^{-11}}{[1.22 \times 10^{-10}t + 0.01]^{\frac{1}{2}}} \quad (9)$$

3. Evolving magnetic field  $B_d = 5 \times 10^{14} \text{G} \cdot e^{-\frac{t}{\tau}}$  with  $\tau = 10000 \text{yrs}$ .

$$P_3^2(t) = P_0^2 + 2K \int_0^t B_d^2(t') dt',$$

the second term

$$\begin{aligned} 2KB_d^2 \int_0^t e^{-2\frac{t'}{\tau}} dt' &= \\ &= -2KB_d^2 \frac{\tau}{2} \int_0^t -\frac{2}{\tau} e^{-2\frac{t'}{\tau}} = \\ &= -2KB_d^2 \frac{\tau}{2} \left( e^{-2\frac{t}{\tau}} \right) \Big|_0^t = \\ &= -\tau KB_d^2 \left( e^{-2\frac{t}{\tau}} - 1 \right) = \\ &= \tau KB_d^2 \left( 1 - e^{-2\frac{t}{\tau}} \right), \end{aligned}$$

therefore

$$P_3^2(t) = (0.01 \text{s}^2) + 19.25 \text{s}^2 \left( 1 - e^{-2\frac{t}{\tau}} \right),$$

thus we get the period

$$P_3(t) = \left[ (0.01 \text{s}^2) + 19.25 \text{s}^2 \left( 1 - e^{-2\frac{t}{\tau}} \right) \right]^{\frac{1}{2}}, \quad (10)$$

the angular velocity

$$\omega_3(t) = 2\pi \left[ (0.01 \text{s}^2) + 19.25 \text{s}^2 \left( 1 - e^{-2\frac{t}{\tau}} \right) \right]^{\frac{1}{2}}, \quad (11)$$

and the period derivative

$$\dot{P}_3(t) = \frac{3.05 \times 10^{-11} e^{-6.33 \dots \times 10^{-12} t}}{\left[ 9.62507 \left( 1 - e^{-6.33 \dots \times 10^{-12} t} \right) + 0.01 \right]^{\frac{1}{2}}} \quad (12)$$

To gain a more comprehensive understanding of the system's evolution, we calculate the relative percentage variations of the period for each case. For instance, for Case 1, the relative variation is given by:

$$\Delta\nu_1(t) = \frac{P_1(t) - P_0}{P_0} \cdot 100, \quad (13)$$

where

$$\Delta P_1(t) = \frac{P_1(t) - P_0}{P_0}$$

is the relative variations. Thus in the same way we get

$$\Delta\nu_2(t) = \frac{P_2(t) - P_0}{P_0} \cdot 100, \quad (14)$$

$$\Delta\nu_3(t) = \frac{P_3(t) - P_0}{P_0} \cdot 100. \quad (15)$$

Representative plots of the period evolution for different magnetic field strengths at an initial period of  $P_0 = 1\text{ms}$  are shown in Figure 1. These plots illustrate the percentage variations of the period (1), the evolution of the period on a logarithmic scale (2), the derivatives of the period functions (3), and the normalized derivatives (4). The plots reveal that a stronger magnetic field ( $B_d = 5 \times 10^{14}\text{G}$ ) results in a faster increase in the period over time compared to the weaker field cases. The normalized derivative plot emphasizes the changes in the period derivative.

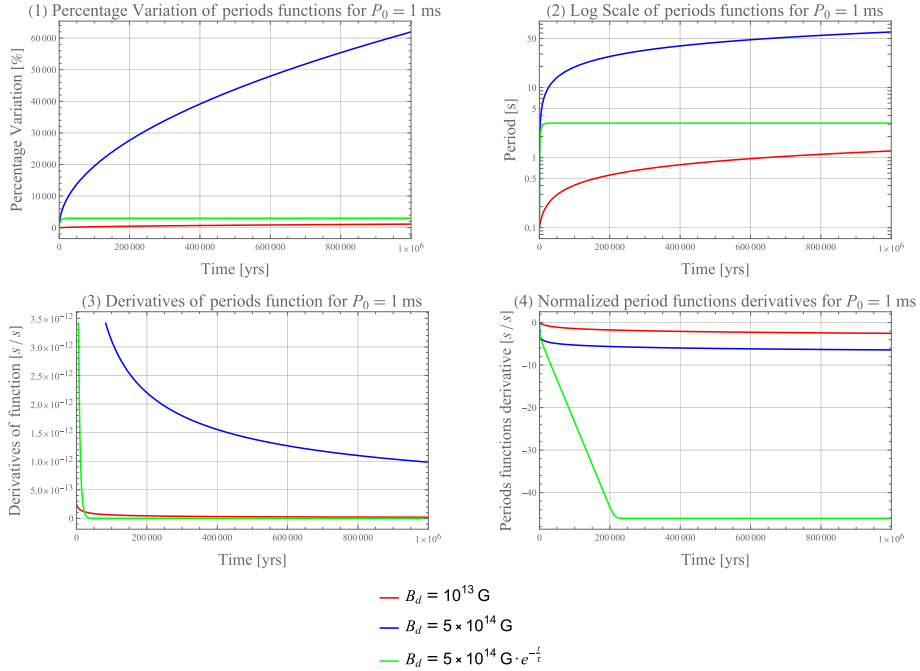


Figure 1: Representatives plots of the evolution of a neutron star with different dipolar component of the surface magnetic field  $B_d$  at initial period  $P_0 = 1\text{ms}$  up to  $1\text{Myr}$ . (1) Visualization of the percentage variation of the period functions, (2) evolution of periods in a logarithmic scale, (3) period functions derivatives and (4) period functions derivatives normalized.

### 3 Comparative analysis of two cases

As in the previous section, we have analyzed the same process with all the same initial condition and for a initial period  $P_0 = 3\text{ms}$ .

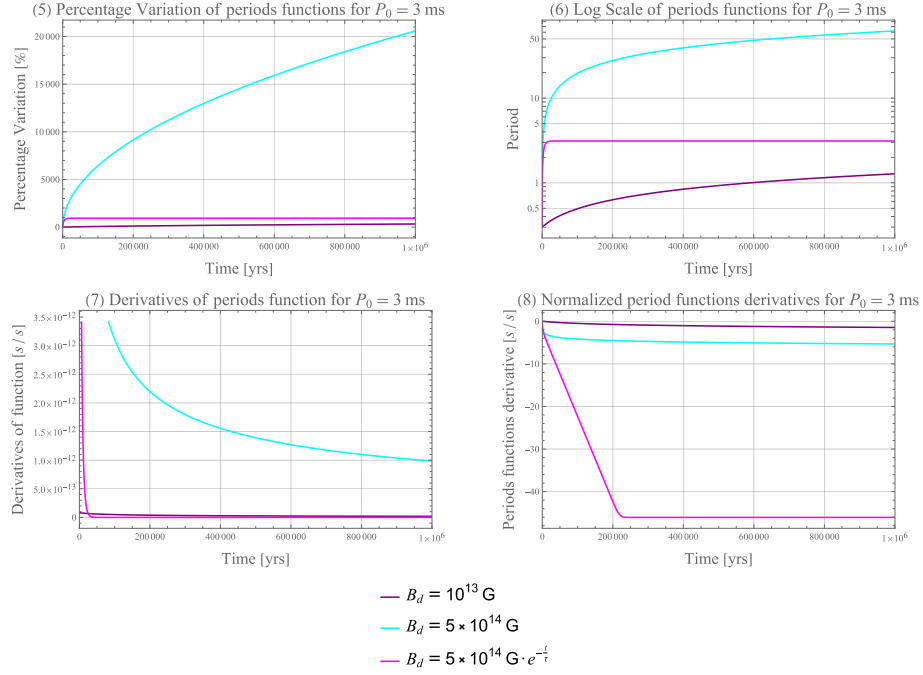


Figure 2: Representatives plots of the evolution of a neutron star with different dipolar component of the surface magnetic field  $B_d$  at initial period  $P_0 = 3$  ms up to 1 Myr. (5) Visualization of the percentage variation of the period functions, (6) evolution of periods in a logarithmic scale, (7) period functions derivatives and (8) period functions derivatives normalized.

It is useful to compare two specific case. Consider the  $B_d = 5 \times 10^{14}$  G. The percentage variation allow to appreciate the different evolutions 3.

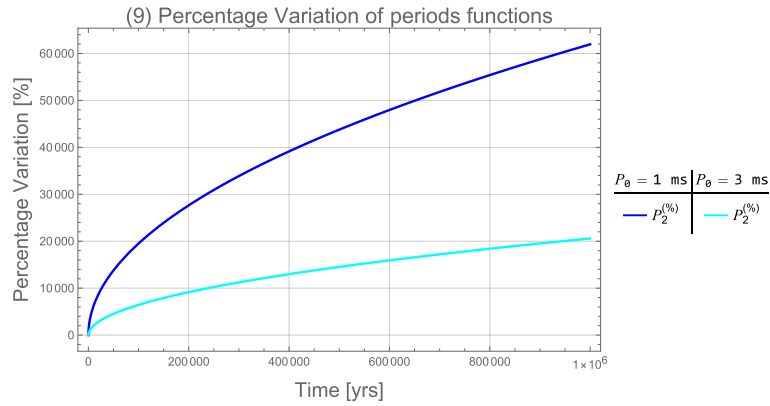


Figure 3: Percentage variation in the period evolution for two initial periods,  $P_0 = 1$  ms and  $P_0 = 3$  ms, under the same magnetic field  $B_d = 5 \times 10^{14}$  G.

The percentage variation increases with time, indicating that the disparity between the cases with different initial periods becomes more pronounced as the system evolves. While both curves show an upward trend, they do not appear to plateau within the observed time frame, suggesting that the influence of the initial period persists over long timescales. The system with an initial period of 1ms (blue) exhibits a more significant deviation compared to the 3ms case (cyan). This behavior implies that systems with shorter initial periods evolve at a faster rate, highlighting the strong impact of the initial spin on the long-term evolution of the system.

## 4 $P - \dot{P}$ diagram and conclusions

Finally, we present the  $P - \dot{P}$  diagram which illustrates the evolution of neutron stars in  $P - \dot{P}$  space. The trajectories show that neutron stars with higher magnetic fields experience a faster spin-down, resulting in steeper trajectories in the diagram.

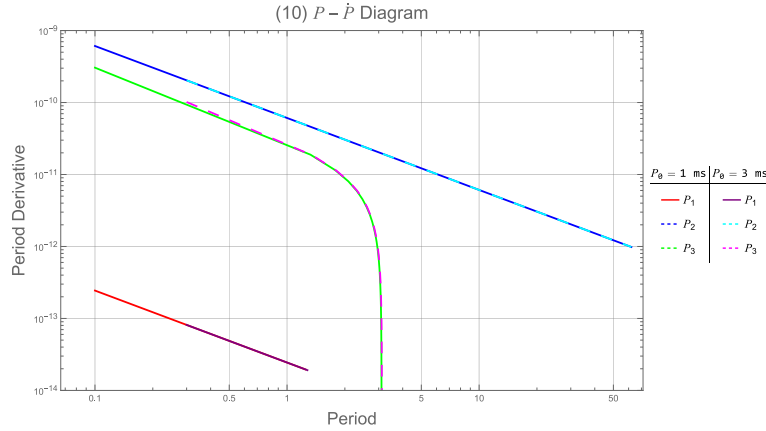


Figure 4:  $P - \dot{P}$  diagram, visualization of all trajectories evolution.

It is also important to note that a cut-off of  $10^{-14}$  is applied. This is justified by the fact that spin-down is an effective theory.

The analysis of the neutron star evolution under different magnetic field conditions demonstrates the significant role of the initial period and magnetic field strength in determining the long-term evolution of the system. The results provide insights into the dynamics of neutron stars and highlight the influence of magnetic braking on their spin-down rates.

## References

- [1] Y. W. J. Lee et al. “The emission of interpulses by a 6.45-h-period coherent radio transient”. In: *Nature Astronomy* (Jan. 2025). ISSN: 2397-3366. DOI: 10.1038/s41550-024-02452-z. URL: <http://dx.doi.org/10.1038/s41550-024-02452-z>.